Lecture 2

Kinematics of Particles I : Rectilinear motion

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MC2103

Fall 2026

Lecture 2-1

https://youtu.be/oZ_DDCO3Itc

Lecture 2-2

Introduction

Kinematic relationships are used to help us determine the trajectory of a Snowboarder completing a jump, the orbital speed of a satellite, and accelerations during acrobatic flying.




Introduction

Dynamics includes

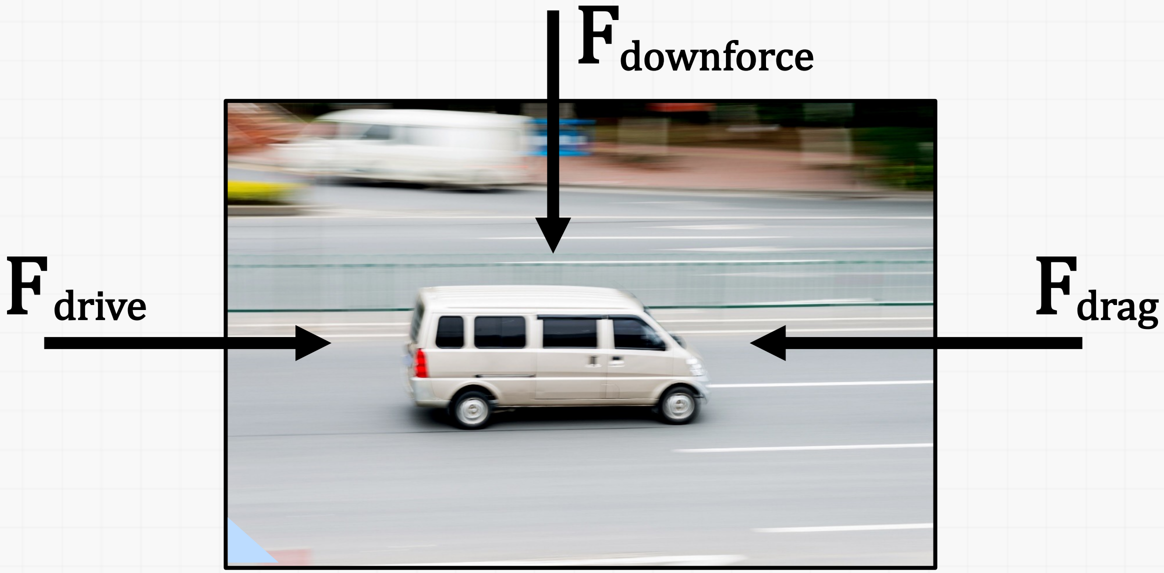
- Kinematics
  - study of the geometry of motion
  - Relates displacement, velocity, acceleration, and time without reference to the cause of motion

GIST

Introduction

Dynamics includes

- Kinetics
  - study of the relations existing between the forces acting on a body, the mass of the body, and the motion of the body.
  - Kinetics is used to predict the motion caused by given forces or to determine the forces required to produce a given motion.



Introduction

Particle kinematics includes

- Rectilinear motion
  - position, velocity, and acceleration of a particle as it moves along a straight line


- Curvilinear motion
  - position, velocity, and acceleration of a particle as it moves along a curved line in two or three dimensions.

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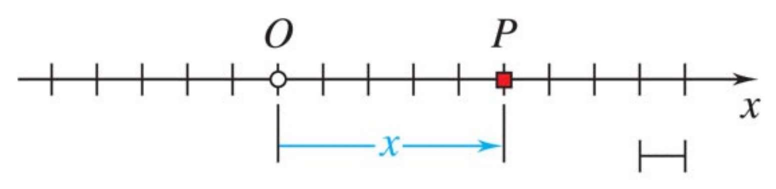
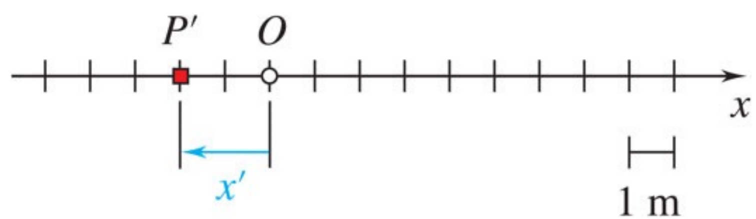
Rectilinear Motion: Position, Velocity & Acceleration

- Rectilinear motion
  - particle moving along a straight line



(a)

- Position coordinate
  - defined by positive or negative distance from a fixed origin on the line.



(b)

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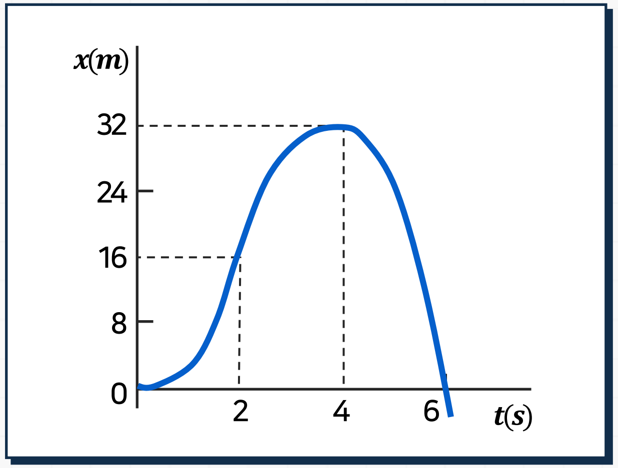
Rectilinear Motion: Position, Velocity & Acceleration

- The motion of a particle is known if the position coordinate for particle is known for every value of time t.
- May be expressed in the form of a function, e.g., $x = 6t^2 - t^3$ or in the form of a graph x vs. t.



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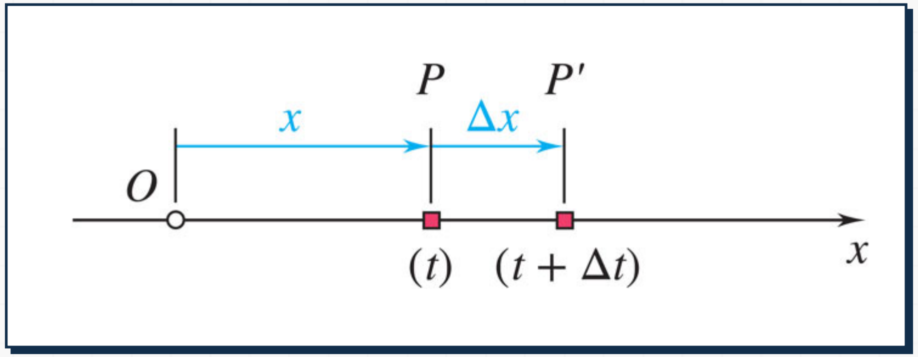
Rectilinear Motion: Position, Velocity & Acceleration

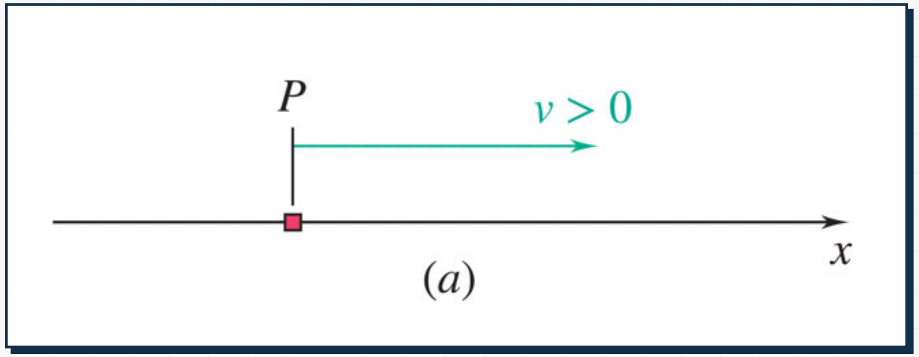
- Consider particle which occupies position P at time t and P' at $t+\Delta t$,

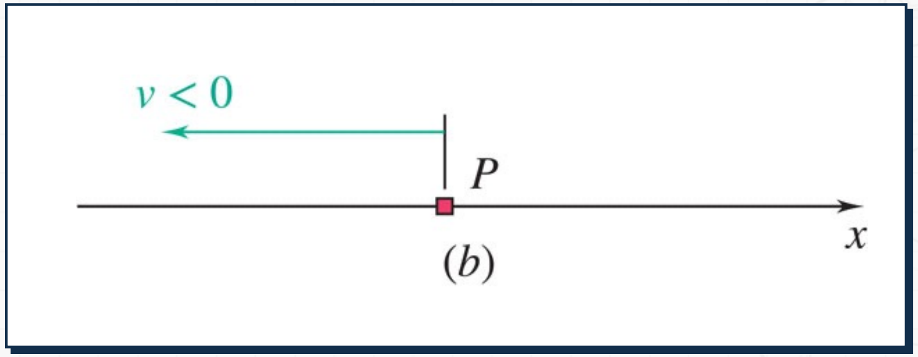


$$Average velocity = \frac{\Delta x}{\Delta t}$$

$$Instantaneous\ velocity = v = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t}$$

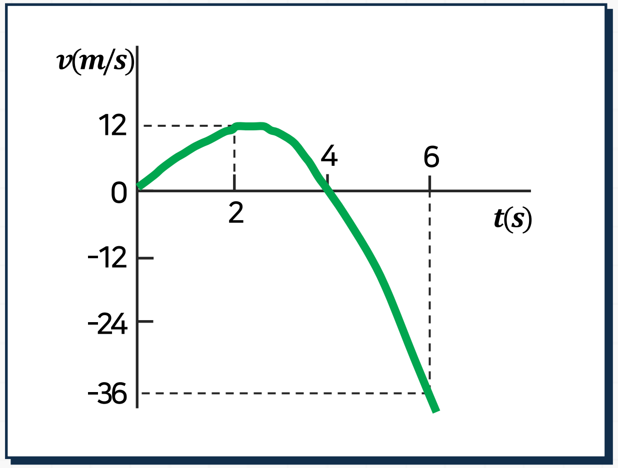
- Instantaneous velocity may be positive or negative.
- Magnitude of velocity is referred to as particle speed.





Rectilinear Motion: Position, Velocity & Acceleration

- From the definition of a derivative,



$$v = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

$$\text{e.g., } x = 6t^2 - t^3$$

$$v = \frac{dx}{dt} = 12t - 3t^2$$

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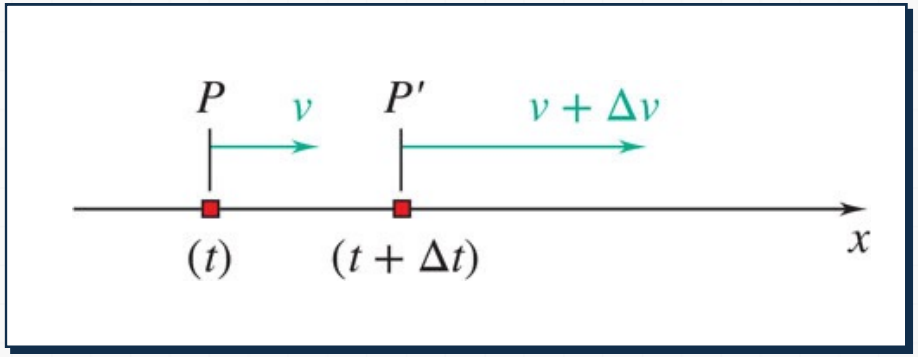
Rectilinear Motion: Position, Velocity & Acceleration

- Consider particle with velocity v at time t and v' at $t+\Delta t$,



$$\text{Instantaneous acceleration} = a = \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t}$$

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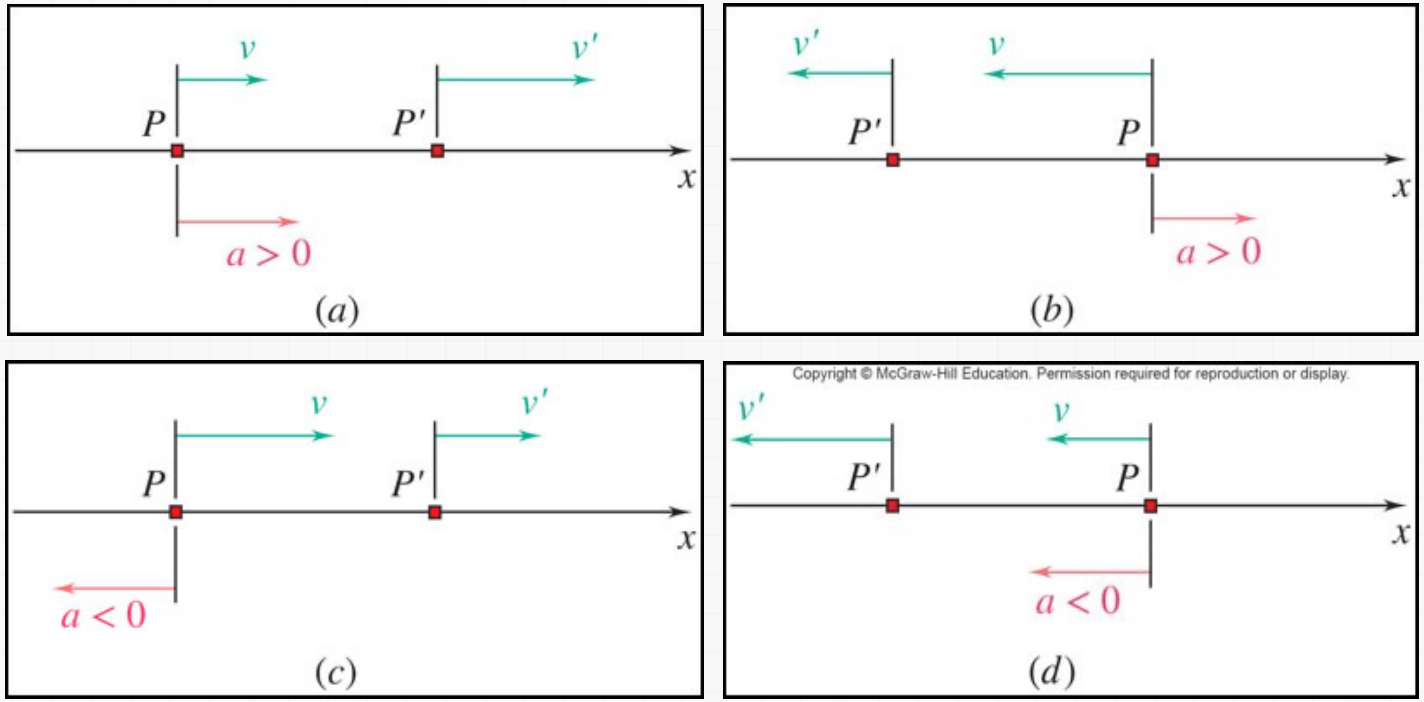
Rectilinear Motion: Position, Velocity & Acceleration

- Instantaneous acceleration may be:
  - positive: increasing positive velocity or decreasing negative velocity
  - negative: decreasing positive velocity or increasing negative velocity



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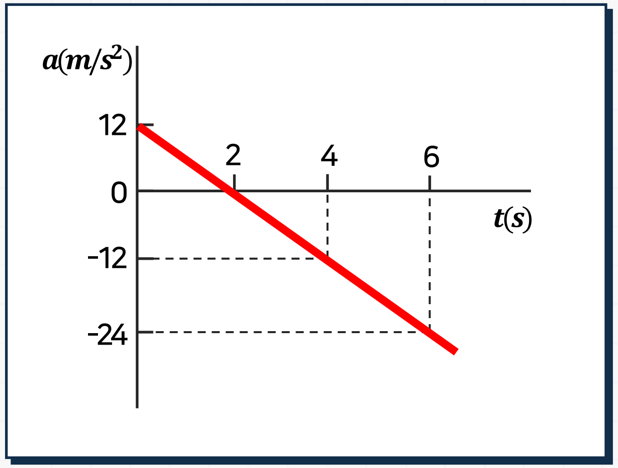
Rectilinear Motion: Position, Velocity & Acceleration

- From the definition of a derivative,



$$a = \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt} = \frac{d^2x}{dt^2}$$

$$\text{e.g., } v = 12t - 3t^2$$

$$a = \frac{dv}{dt} = 12 - 6t$$

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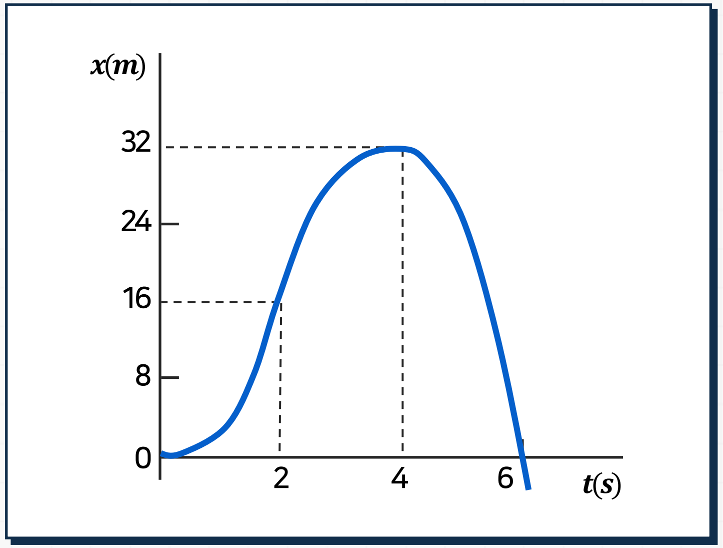
Rectilinear Motion: Position, Velocity & Acceleration



From our example

$$x = 6t^2 - t^3$$

$$v = \frac{dx}{dt} = 12t - 3t^2$$

$$a = \frac{dv}{dt} = \frac{d^2x}{dt^2} = 12 - 6t$$

- What are x, v, and a at $t = 2\text{s}$?

at $t = 2\text{s}$, $x = 16\text{m}$, $v = v_{max} = 12\text{m/s}$, $a = 0\text{m/s}^2$

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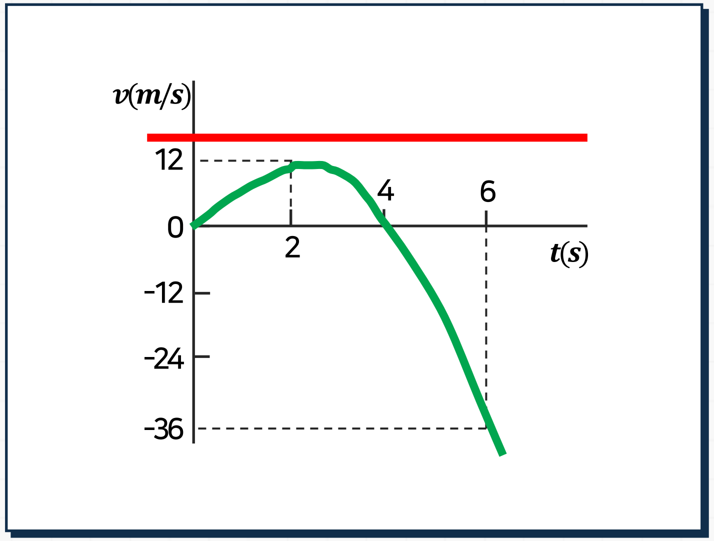
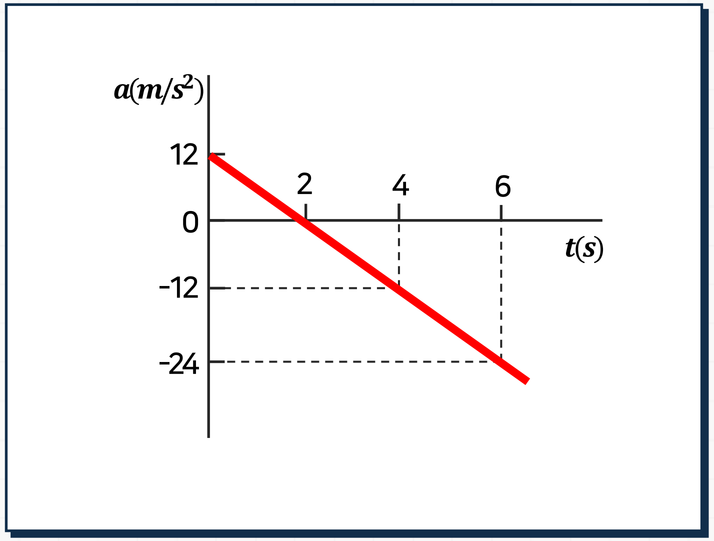
Rectilinear Motion: Position, Velocity & Acceleration





- Note that vmax occurs when a = 0, and that the slope of the velocity curve is zero at this point. What are x, v, and a at $t = 4\text{s}$?

$$\text{at } t = 4\text{s},\ x = x_{max} = 32\text{m},\ v = 0\text{m/s},\ a = -12\text{m/s}^2$$

Determining the Motion of a Particle

We often determine accelerations from the forces applied (kinetics will be covered later)
Generally have three classes of motion

acceleration given as a function of time, $a = f(t)$.

acceleration given as a function of position, $a = f(x)$.

acceleration given as a function of velocity, $a = f(v)$.

Determining the Motion of a Particle

Can you think of a physical example of when force is a function of position?
When force is a function of velocity?

A Spring

Drag

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Acceleration as a function of time, position, or velocity

If…	Kinematic relationship	Integrate
$a=a(t)$	$\frac{dv}{dt}=a(t)$	$\int_{v_0}^{v} dv = \int_{0}^{t} a(t)dt$
$a=a(x)$	$dt=\frac{dx}{v}$ and $a=\frac{dv}{dt}$ ↓ $v\,dv = a(x)dx$	$\int_{v_0}^{v} vdv = \int_{x_0}^{v} a(x)dx$
$a=a(v)$	$\frac{dv}{dt}=a(v)$	$\int_{v_0}^{v} \frac{dv}{a(v)} = \int_{0}^{t} dt$
	$v\frac{dv}{dx}=a(v)$	$\int_{x_0}^{x} dx = \int_{v_0}^{v} \frac{vdv}{a(v)}$

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Uniform Rectilinear Motion

Once a safe speed of descent for a vertical landing is reached, a Harrier jet pilot will adjust the vertical thrusters to equal the weight of the aircraft. The plane then travels at a constant velocity downward.
If motion is in a straight line, this is uniform rectilinear motion.

Uniform Rectilinear Motion

For a particle in uniform rectilinear motion, the acceleration is zero and the velocity is constant.

$$\frac{dx}{dt} = v = constant \qquad \int_{x_0}^{x} dx = v \int_{0}^{t} dt \qquad \begin{aligned} x &= x_0 - vt \\ x &= x_0 + vt \end{aligned}$$

Careful – these only apply to uniform rectilinear motion!

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Uniformly Accelerated Rectilinear Motion

For a particle in uniformly accelerated rectilinear motion, the acceleration of the particle is constant. You may recognize these constant acceleration equations from your physics courses.

$$\frac{dv}{dt} = a = constant \qquad \int_{v_0}^{v} dv = a\int_{0}^{t} dt \qquad v = v_0 + at$$

$$\frac{dx}{dt} = v_0 + at \qquad \int_{x_0}^{x} dx = \int_{0}^{t} (v_0 + at)dt \qquad x = x_0 + v_0 t + \frac{1}{2}at^2$$

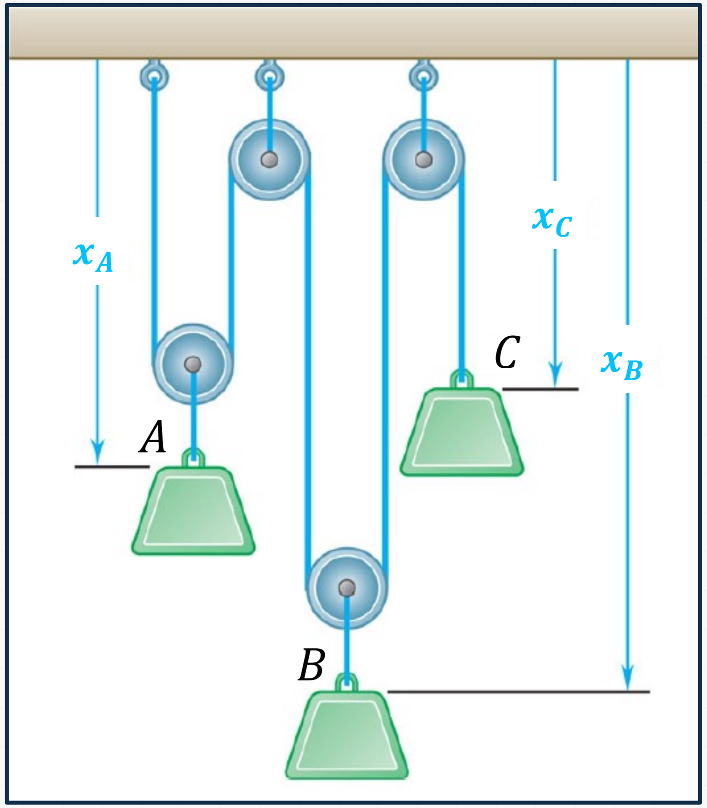
$$v\frac{dv}{dx} = a = constant \qquad \int_{v_0}^{v} vdv = a\int_{x_0}^{x} dx \qquad v^2 = v_0^2 + 2a(x - x_0)$$

Careful – these only apply to uniform rectilinear motion!

Lecture 2-3

Motion of Several Particles

We may be interested in the motion of several different particles, whose motion may be independent or linked together.



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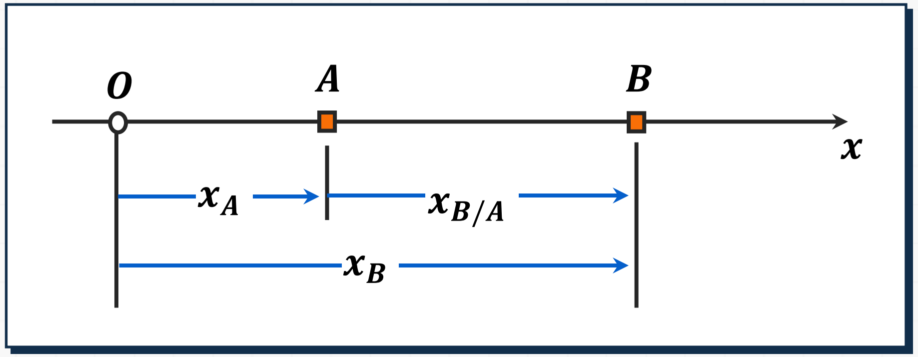
Motion of Several Particles: Relative Motion

For particles moving along the same line, time should be recorded from the same starting instant and displacements should be measured from the same origin in the same direction.



$$x_{B/A} = x_B - x_A = \text{ relative position of } B \text{ with respect to } A$$

$$x_B = x_A + x_{B/A}$$

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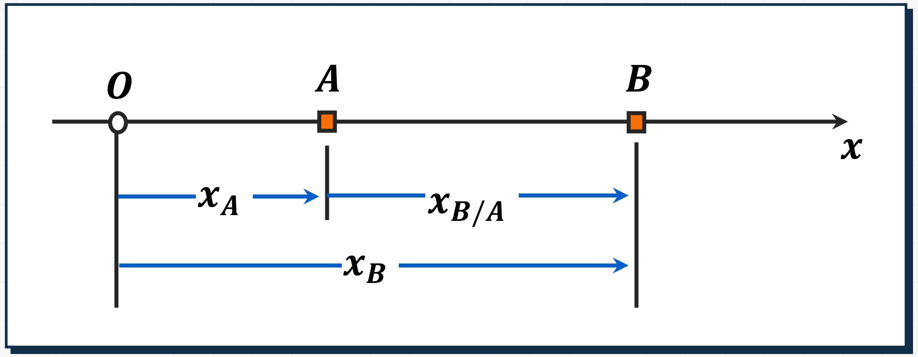
Motion of Several Particles: Relative Motion

For particles moving along the same line, time should be recorded from the same starting instant and displacements should be measured from the same origin in the same direction.

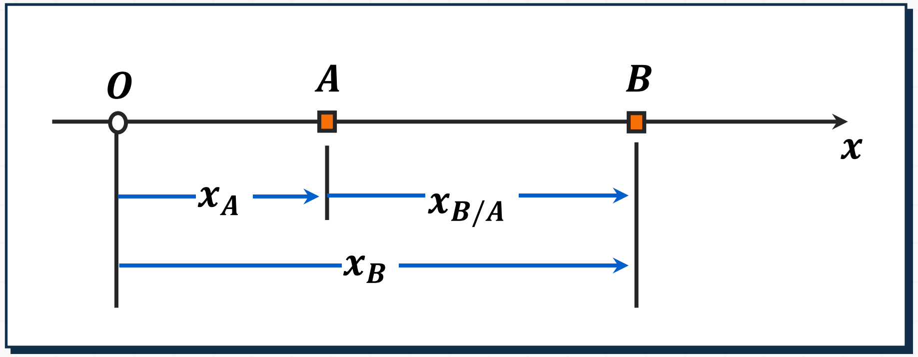


$v_{B/A} = v_B - v_A =$ relative velocity of B with respect to A

$v_B = v_A + v_{B/A}$

Motion of Several Particles: Relative Motion

For particles moving along the same line, time should be recorded from the same starting instant and displacements should be measured from the same origin in the same direction.

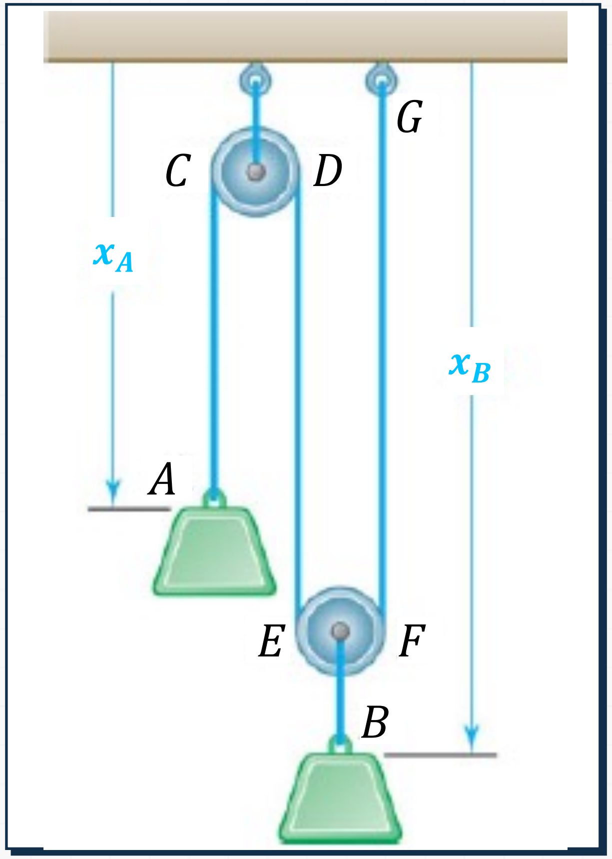


$$a_{B/A} = a_B - a_A = \text{relative acceleration of } B \text{ with respect to } A$$

$$a_B = a_A + a_{B/A}$$

Motion of Several Particles: Dependent Motion

Position of two blocks



- Position of a particle may depend on position of one or more other particles.
- Position of block B depends on position of block A.
  - Since rope is of constant length, it follows that sum of lengths of segments must be constant.

$$x_A + 2x_B = constant\ (one\ degree\ of\ freedome)$$

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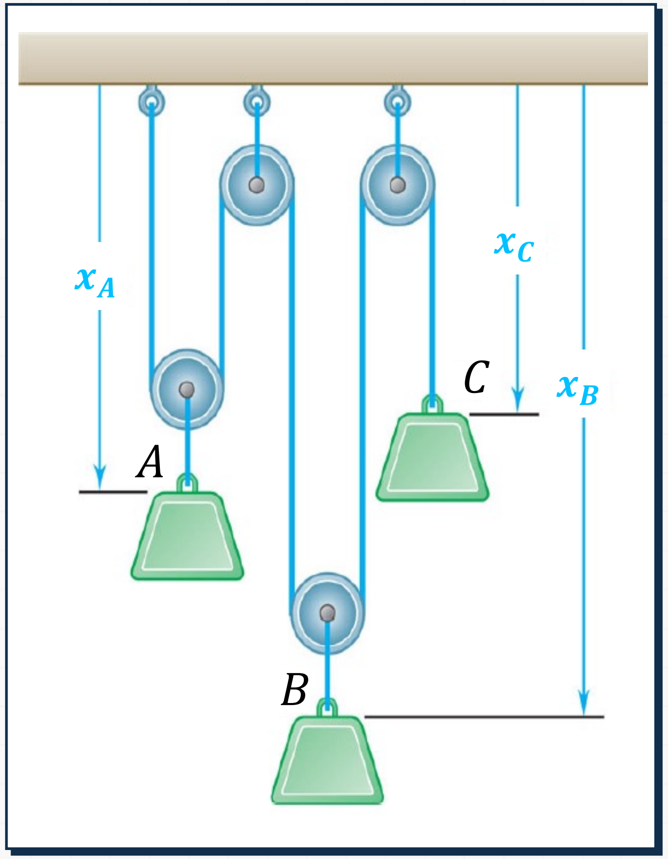
Motion of Several Particles: Dependent Motion

Position of three blocks



- Position of three blocks are dependent.

$$2x_A + 2x_B + x_c = constant\ (two\ degrees\ of\ freedome)$$

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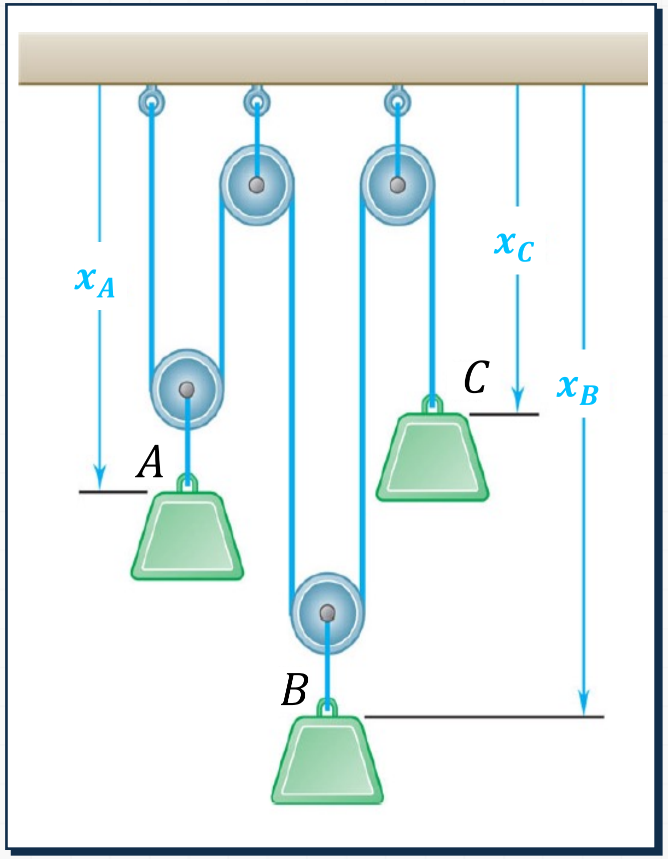
Motion of Several Particles: Dependent Motion

Position of three blocks



- For linearly related positions, similar relations hold between velocities and accelerations.

$$2\frac{dx_A}{dt} + 2\frac{dx_B}{dt} + 2\frac{dx_C}{dt} = 0 \quad \text{or} \quad 2v_A + 2v_B + v_c = 0$$

$$2\frac{dv_A}{dt} + 2\frac{dv_B}{dt} + 2\frac{dv_C}{dt} = 0 \quad \text{or} \quad 2a_A + 2a_B + a_c = 0$$

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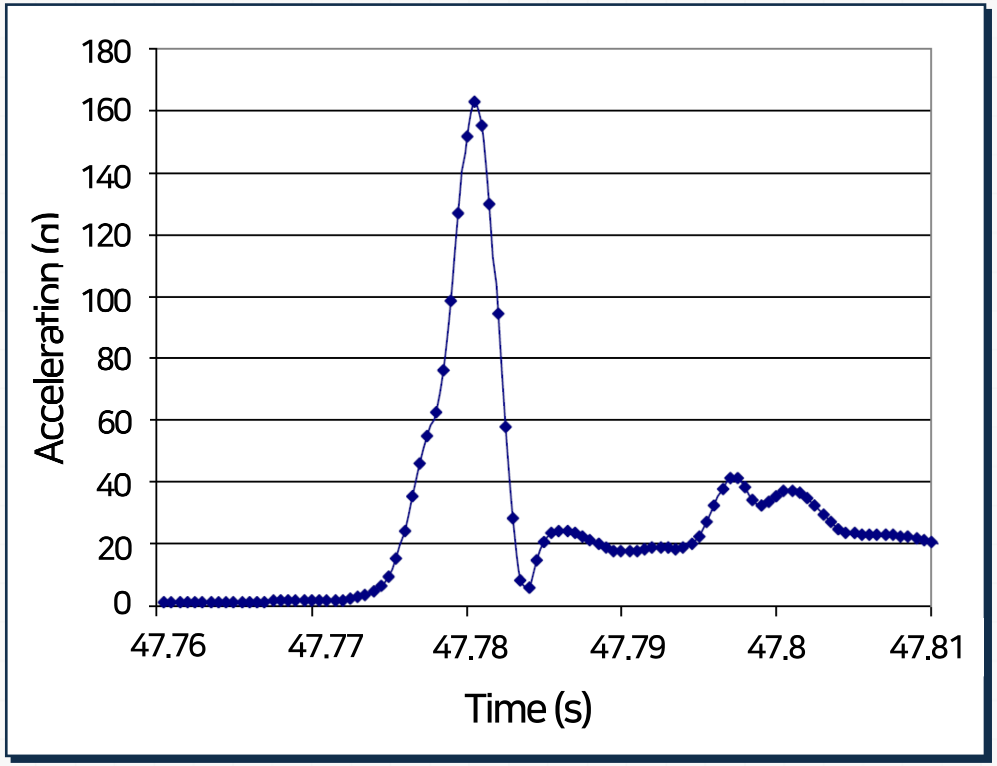
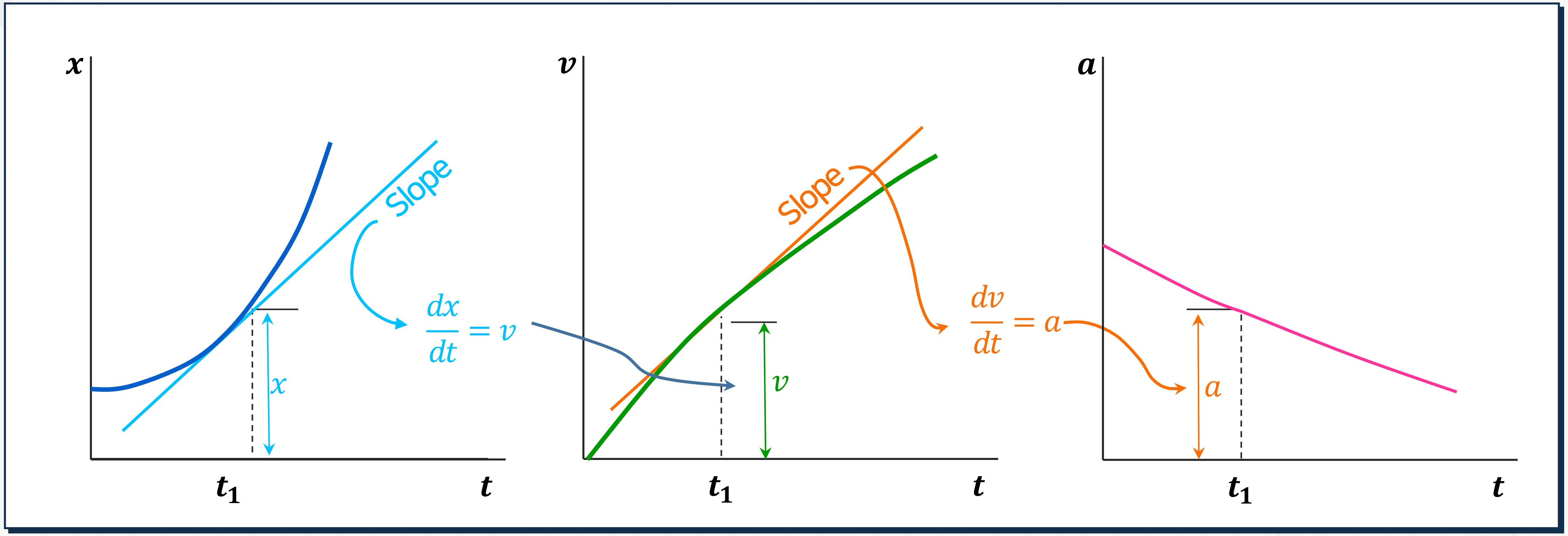
Graphical Solutions 1

Engineers often collect position, velocity, and acceleration data. Graphical solutions are often useful in analyzing these data.



Acceleration data from a head impact during a round of boxing.

Graphical Solutions 2



- Given the $x - t$ curve, the $v - t$ curve is equal to the $x - t$ curve slope.
- Given the $v - t$ curve, the $a - t$ curve is equal to the $v - t$ curve slope.

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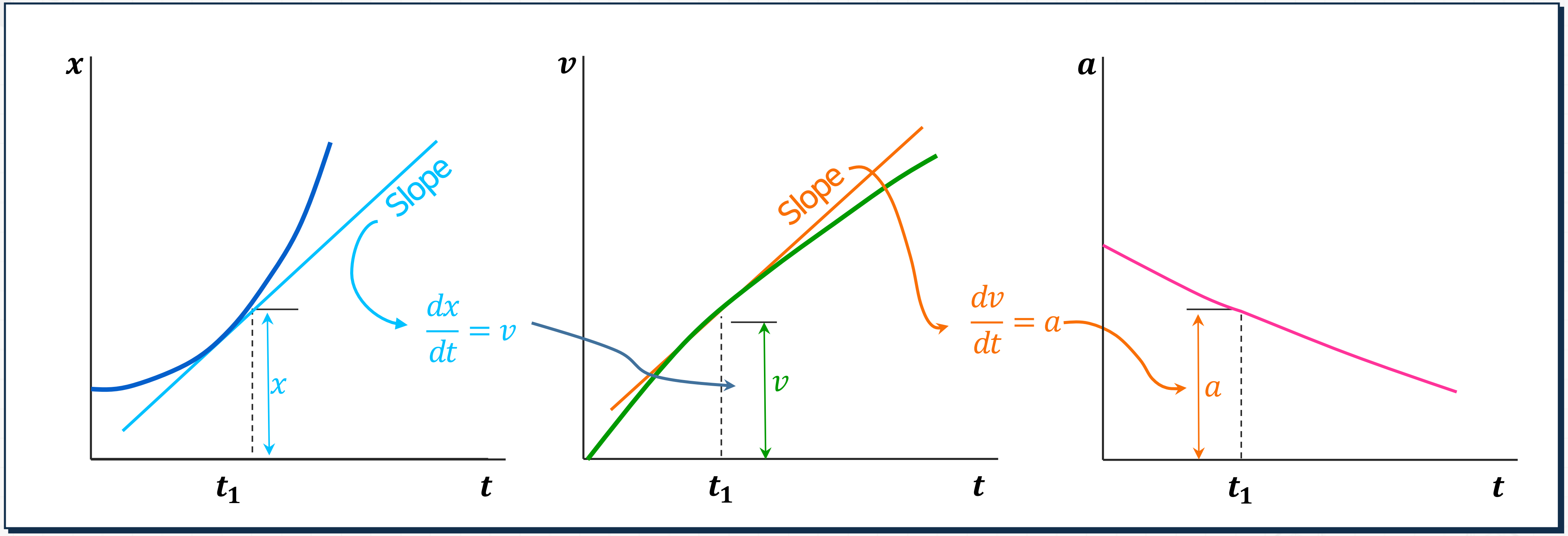
Graphical Solutions 3

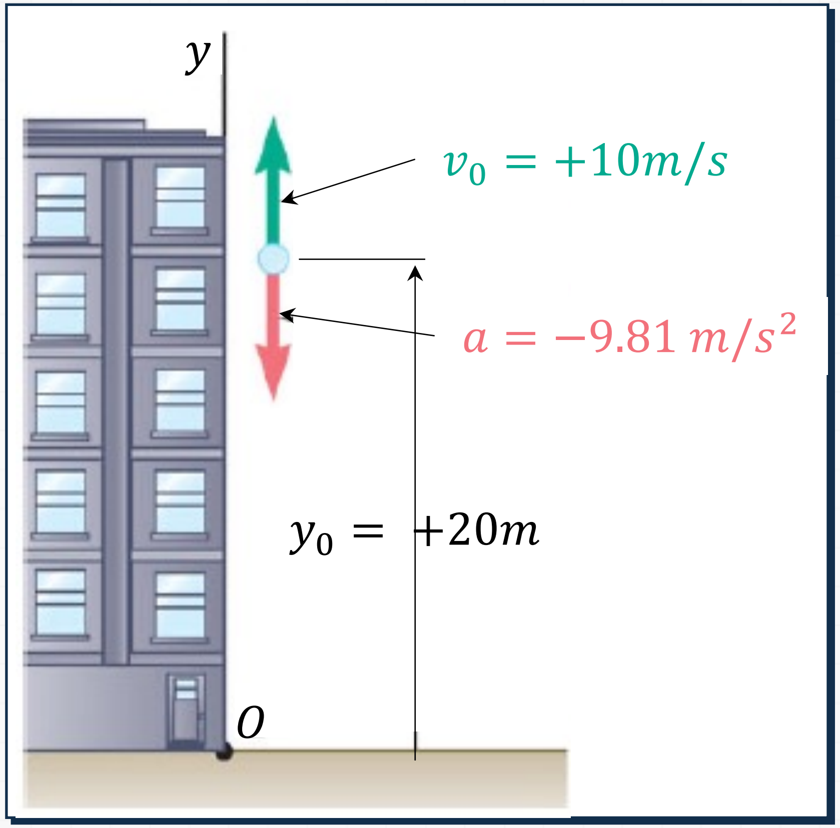


- Given the $a - t$ curve, the change in velocity between t_1 and t_2 is equal to the area under the a-t curve between t_1 and t_2.
- Given the $v - t$ curve, the change in position between t_1 and t_2 is equal to the area under the $v - t$ curve between t_1 and t_2.

Week 2-4

Sample Problem 1

Ball tossed with 10m/s vertical velocity from window 20m above ground.



- Determine
  - Velocity and elevation above ground at time t
  - Highest elevation reached by ball and corresponding time
  - Time when ball will hit the ground
  - and corresponding velocity

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Sample Problem 1

Ball tossed with 10m/s vertical velocity from window 20m above ground.

- Strategy
  - Acceleration is constant, so we can directly integrate twice to find $v(t)$ and $y(t)$.
  - Solve for t when velocity equals zero(time for maximum elevation) and evaluate corresponding altitude.
  - Solve for t when altitude equals zero(time for ground impact) and evaluate corresponding velocity.

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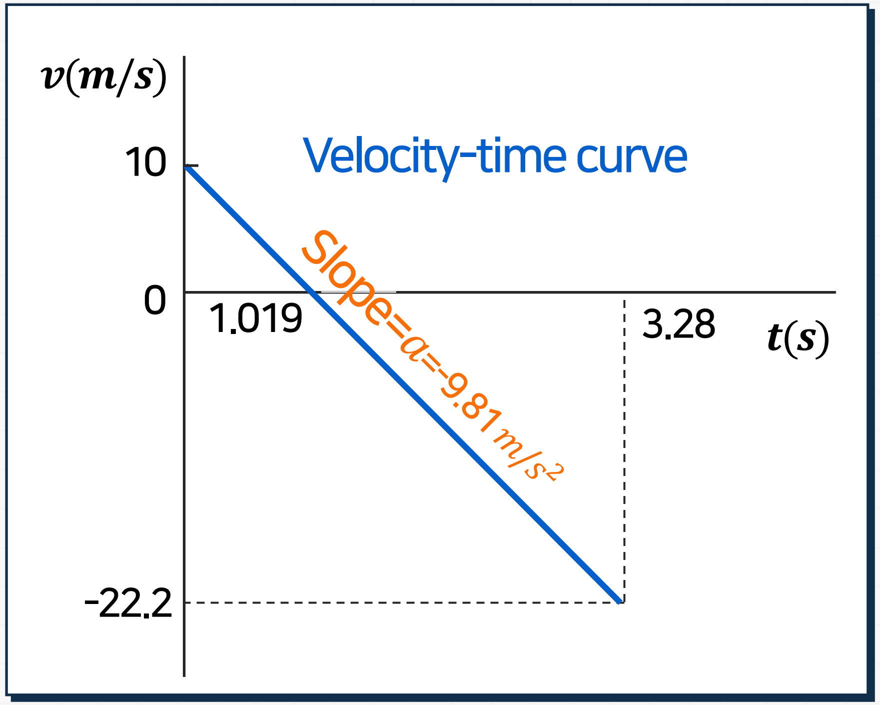
Sample Problem 1

Modeling and Analysis

- Integrate twice to find $v(t)$ and $y(t)$.



$$\frac{dv}{dt} = a = -9.81\, m/s^2$$

$$\int_{v_0}^{v(t)} dv = -\int_0^t 9.81 dt \qquad v(t) - v_0 = -9.81t$$

$$v(t) = 10 - 9.81t \text{ (m/s)}$$

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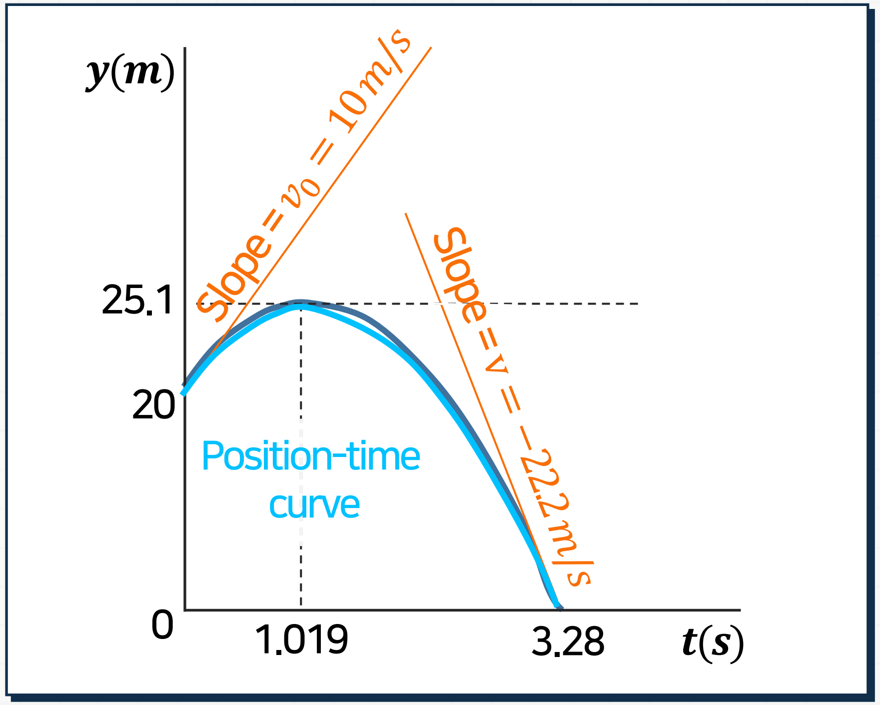
Sample Problem 1

Modeling and Analysis

- Integrate twice to find $v(t)$ and $y(t)$.



$$\frac{dy}{dt} = v = 10 - 9.81t$$

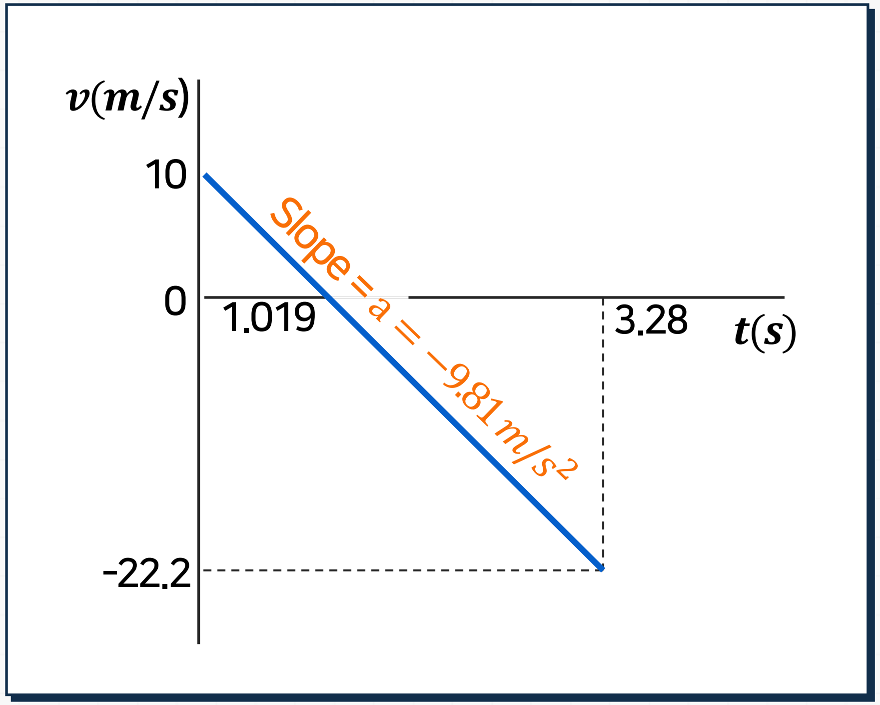
$$\int_{y_0}^{y(t)} dy = \int_0^t (10 - 9.81t)dt \quad y(t) - y_0 = 10t - \frac{1}{2}9.81t^2$$

$$\boldsymbol{y(t) = 20 + (10)t - (4.905)t^2\ m}$$

Sample Problem 1

Modeling and Analysis

- To find the highest elevation reached, first solve for t when velocity equals zero.



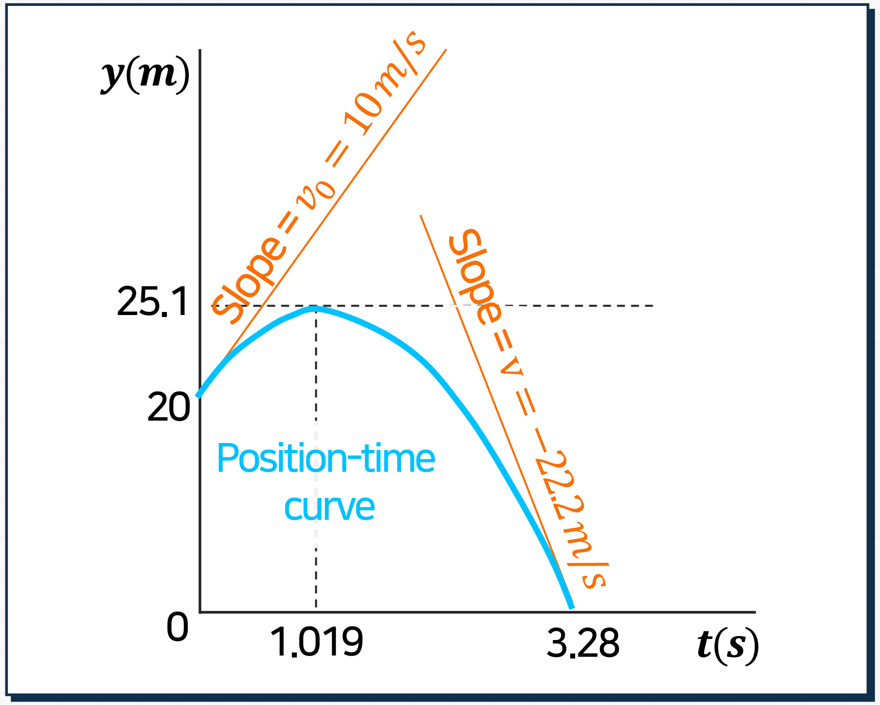
$$v(t) = 10 - (9.81)t = 0$$

$$t = 1.019\ \mathrm{s}$$

Sample Problem 1

Modeling and Analysis

- Now evaluate the altitude at the time corresponding to zero vertical velocity.



$$y(t) = 20 - (10)t - (4.905)t^2 \ m$$

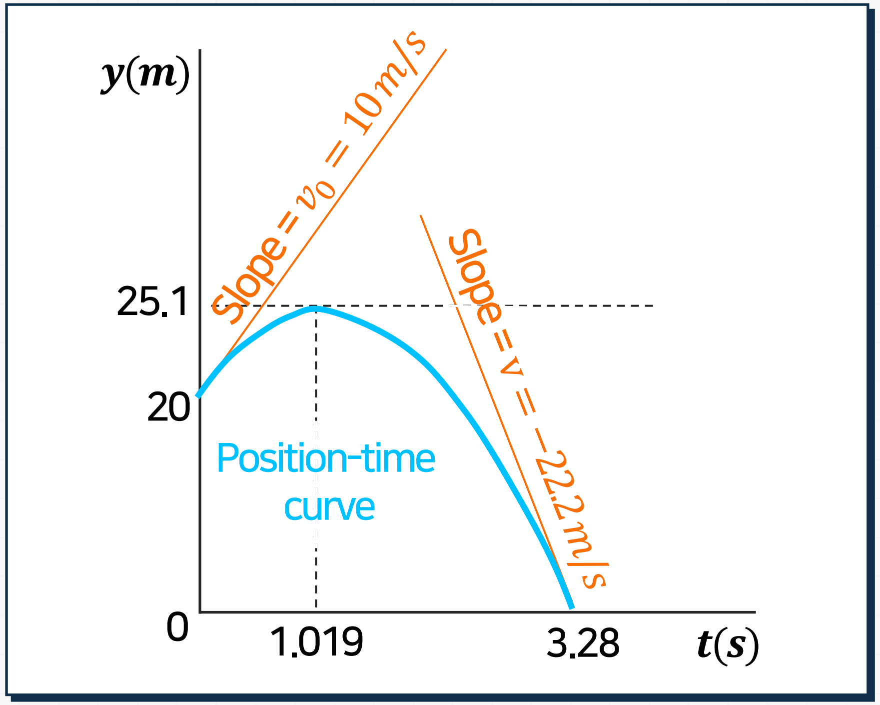
$$y = 20 + (10)(1.019) - (4.905)(1.019)^2 \ m$$

$$\boxed{y = 25.1 \ m}$$

Sample Problem 1

Modeling and Analysis

- To find the velocity when the ball hits the ground, first solve for t when altitude equals zero, and then evaluate the velocity at that time.



$$y(t) = 20 + (10)t - (4.905)t^2 = 0$$

$$t = -1.243s \ (meaningless)$$
$$t = 3.28s$$

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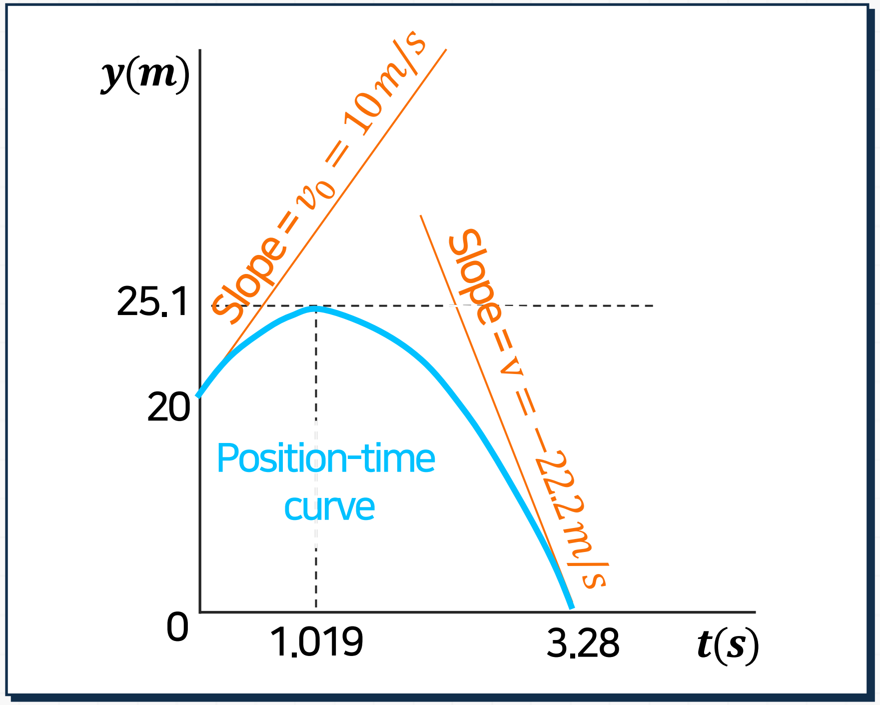
Sample Problem 1

Modeling and Analysis

- To find the velocity when the ball hits the ground, first solve for t when altitude equals zero, and then evaluate the velocity at that time.



$$v(t) = 10 - (9.81)t$$

$$v(3.28) = 10 - (9.81)(3.28)$$

$$\boldsymbol{v = -22.2}\ \text{(m/s)}$$

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Sample Problem 1

Reflect and Think

When the acceleration is constant, the velocity changes linearly, and the position is a quadratic function of time.

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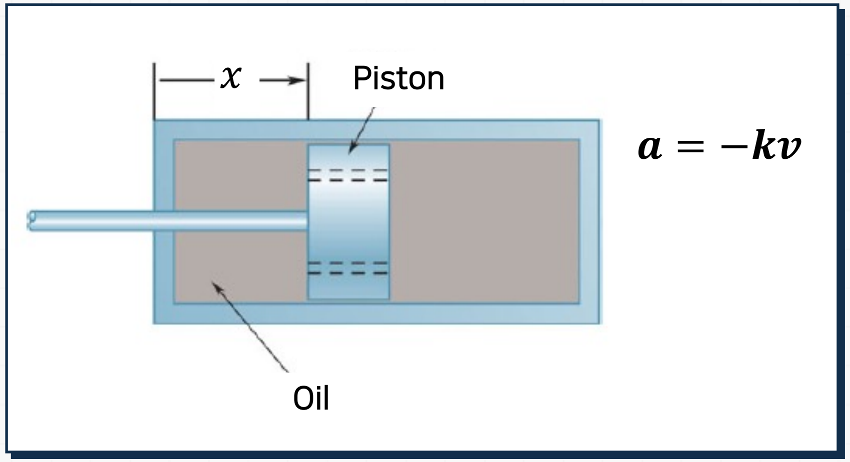
Sample Problem 2

A mountain bike shock mechanism used to provide shock absorption consists of a piston that travels in an oil-filled cylinder.

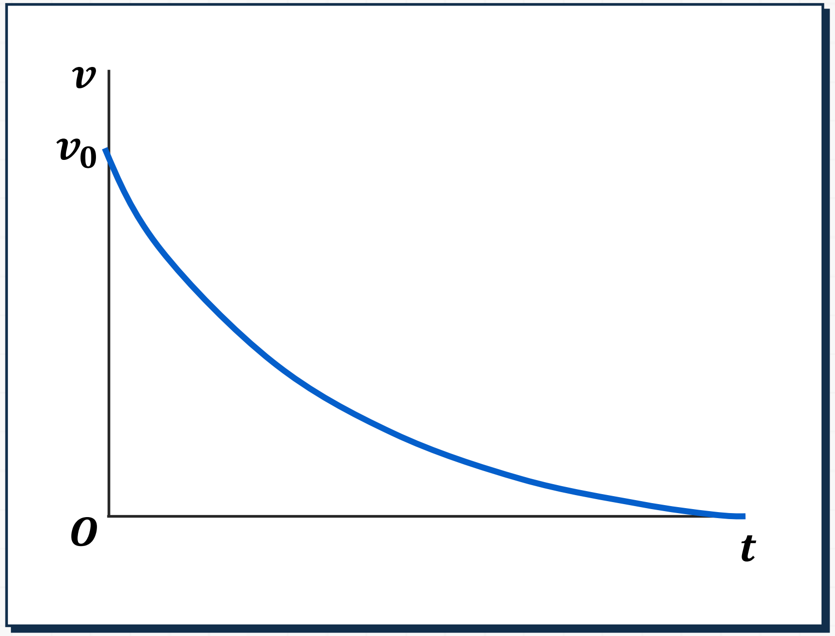


- Determine
  - As the cylinder is given an initial velocity V_0, the piston moves and oil is forced through orifices in piston, causing piston and cylinder to decelerate at rate proportional to their velocity.
  - Determine $v(t)$, $x(t)$, and $v(x)$.

Sample Problem 2

Modeling and Analysis

- Integrate $a = \frac{dv}{dt} = -kv$ to find $v(t)$.



$$a = \frac{dv}{dt} = -kv$$

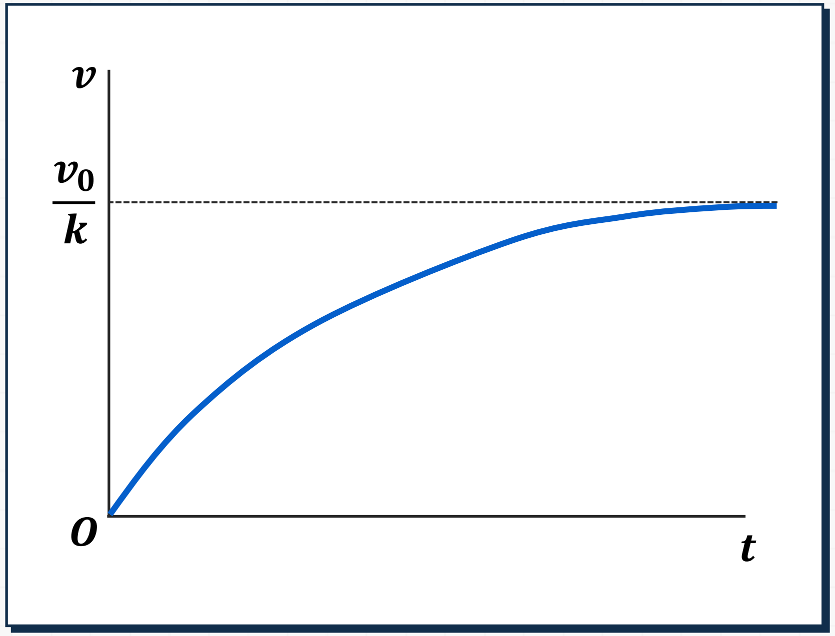
$$\int_{v_0}^{v} \frac{dv}{v} = -k \int_0^t dt \qquad ln\frac{v(t)}{v_0} = -kt$$

$$\boxed{v(t) = v_0 e^{-kt}}$$

Sample Problem 2

Modeling and Analysis

- Integrate $v(t) = \frac{dx}{dt}$ to find $x(t)$.



$$v(t) = \frac{dx}{dt} = v_0 e^{-kt}$$

$$\int_0^x dx = v_0 \int_0^t e^{-kt}\, dt \qquad x(t) = v_0 \left[-\frac{1}{k} e^{-kt} \right]_0^t$$

$$\boxed{x(t) = \frac{v_0}{k}\left(1 - e^{-kt}\right)}$$

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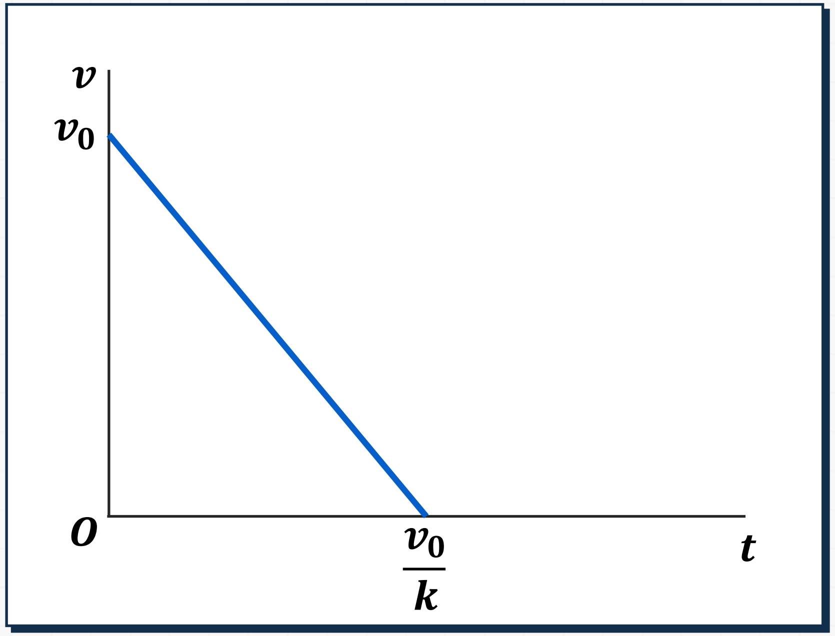
Sample Problem 2

Modeling and Analysis

- Integrate $a = v\dfrac{dv}{dx} = -kv$ to find $v(x)$.



$a = v\dfrac{dv}{dx} = -kv$ to find $v(x)$

$$a = v\frac{dv}{dx} = -kv \qquad dv = -kdx \qquad \int_{v_0}^{v} dv = -k\int_{0}^{x} dx$$

$$v - v_0 = -kv$$

$$v = v_0 - kx$$

Sample Problem 2

Reflect and Think

You could have solved part c by eliminating t from the answers obtained for parts a and b. You could use this alternative method as a check. From part a, you obtain $e^{-kt} = v/v_0$; substituting into the answer of part b, you have:

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Sample Problem 2

Reflect and Think

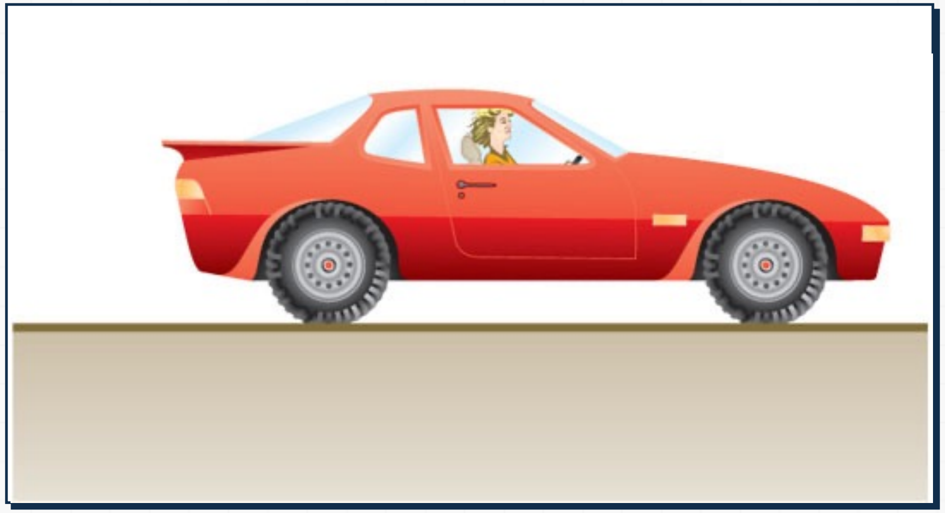
$$x = \frac{v_0}{k}\left(1 - e^{-kt}\right) = \frac{v_0}{k}\left(1 - \frac{v}{v_0}\right)$$

$$v = v_0 - kv$$

$$(checks)$$

Group Problem Solving

The car starts from rest and accelerates according to the relationship



$$a = 3 - 0.001v^2$$

- It travels around a circular track that has a radius of 200 meters.
- Calculate the velocity of the car after it has travelled halfway around the track.
- What is the car's maximum possible speed?

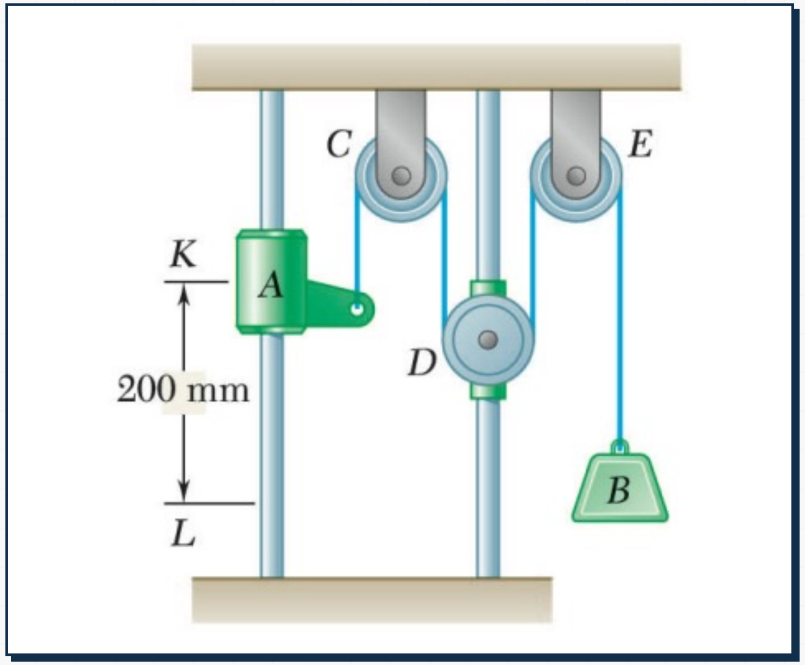
Group Problem Solving

The car starts from rest and accelerates according to the relationship

- Strategy
  - Determine the proper kinematic relationship to apply (is acceleration a function of time, velocity, or position?
  - Determine the total distance the car travels in one-half lap
  - Integrate to determine the velocity after one-half lap

Sample Problem 3

Pulley D is attached to a collar which is pulled down at 75 mm/s.



- At t=0, collar A starts moving down from K with constant acceleration and zero initial velocity.
- Knowing that velocity of collar A is 300 mm/s as it passes L, determine the change in elevation, velocity, and acceleration of block B when block A is at L.

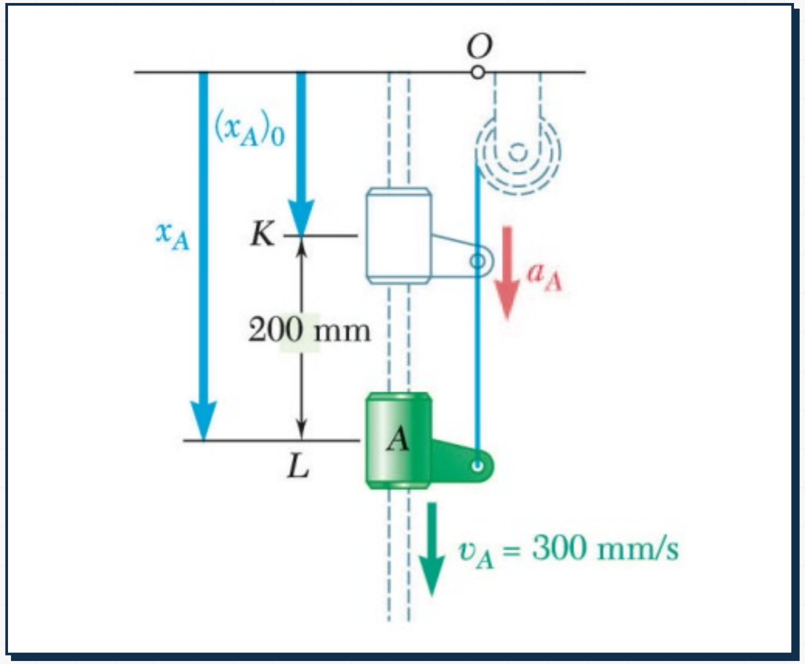
Sample Problem 3

- **Pulley D is attached to a collar which is pulled down at 75 mm/s.**
  - Strategy
    - Define origin at upper horizontal surface with positive displacement downward.
    - Collar A has uniformly accelerated rectilinear motion. Solve for acceleration and time t to reach L .
    - Pulley D has uniform rectilinear motion. Calculate change of position at time t.
    - Block B motion is dependent on motions of collar A and pulley D. Write motion relationship and solve for change of block B position at time t.
    - Differentiate motion relation twice to develop equations for velocity and acceleration of block B.

GIST

Sample Problem 3

Modeling and Analysis

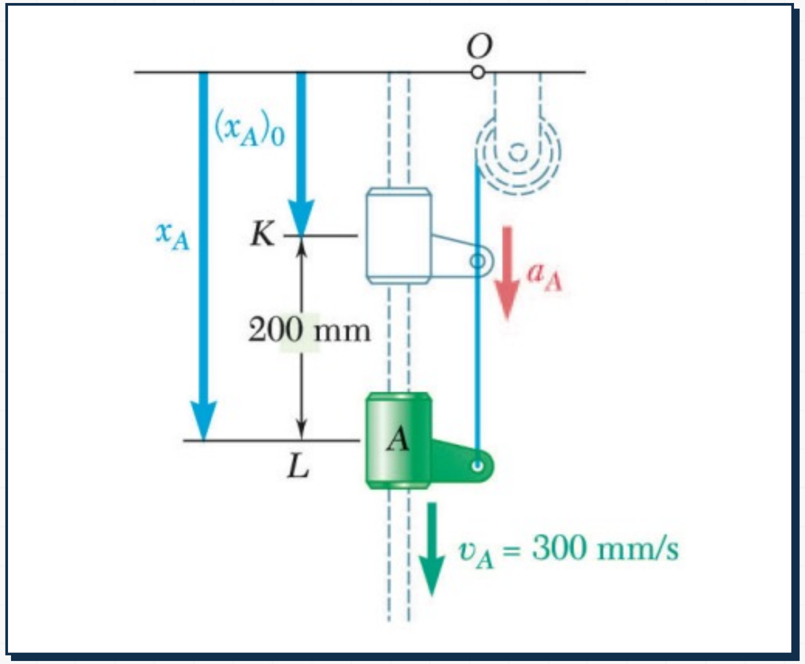


- Define origin at upper horizontal surface with positive displacement downward.
- Collar A has uniformly accelerated rectilinear motion. Solve for acceleration and time t to reach L.

GIST

Sample Problem 3

Modeling and Analysis



$$v_A^2 = (v_A)_0^2 + 2a_A[x_A - (x_A)_0]$$

$$(300)^2 = 2a_A\,(200)$$

$$a_A = 225\ (mm/s^2)$$

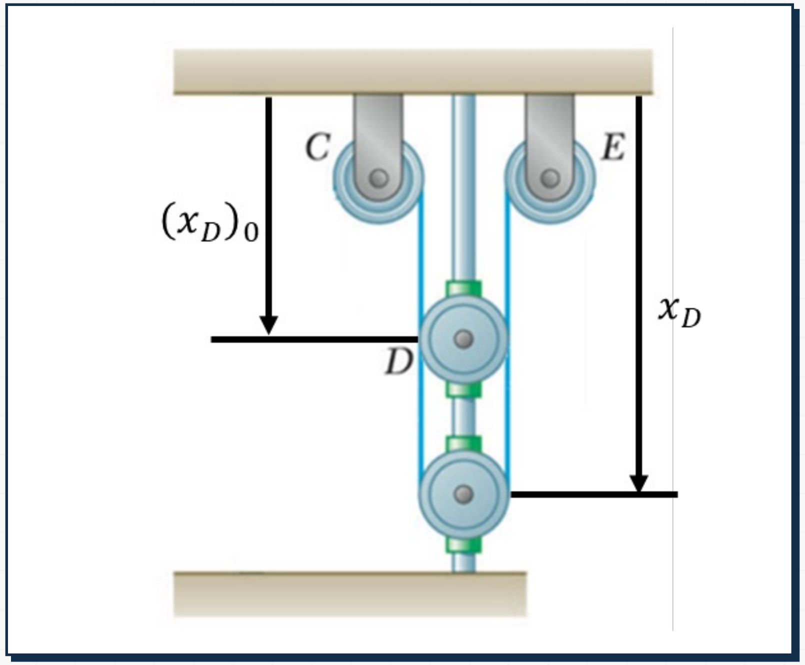
$$v_A = (v_A)_0 + a_A t$$

$$300 = 225t$$

$$t = 1.333\text{s}$$

GIST

Sample Problem 3



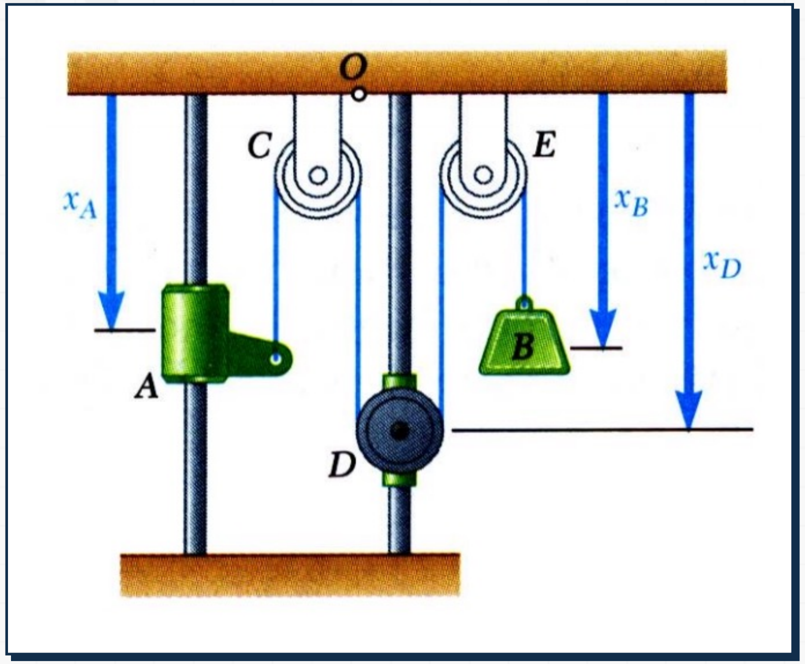
- Pulley D has uniform rectilinear motion. Calculate change of position at time t.

$$x_D = (x_D)_0 + v_D t$$

$$x_D - (x_D)_0 = (75)(1.333s) = 100\ mm$$

GIST

Sample Problem 3



- Block B motion is dependent on motions of collar A and pulley D. Write motion relationship and solve for change of block B position at time t.

Total length of cable remains constant,

$$x_A + 2x_D + x_B = (x_A)_0 + 2(x_D)_0 + (x_B)_0$$

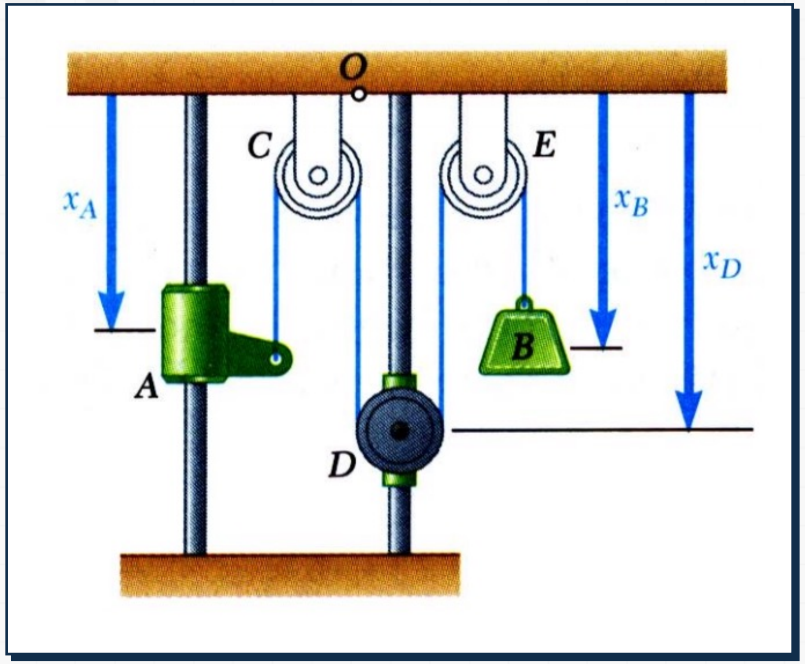
$$[x_A - (x_A)_0] + 2[x_D - (x_D)_0] + [x_B - (x_B)_0] = 0$$

$$(200) + 2(100) + [x_B - (x_B)_0] = 0$$

$$x_B - (x_B)_0 = -400\ mm$$

GIST

Sample Problem 3



- Differentiate motion relation twice to develop equations for velocity and acceleration of block B.

$$x_A + 2x_D + x_B = \text{constant}$$

$$v_A + 2v_D + v_B = 0$$

$$(300) + 2(75) + v_B = 0$$

$$v_B = 450\ mm/s$$

$$a_A + 2a_D + a_B = 0$$

$$(225) + a_B = 0$$

$$a_B = -225\ mm/s^2$$

GIST

Sample Problem 3

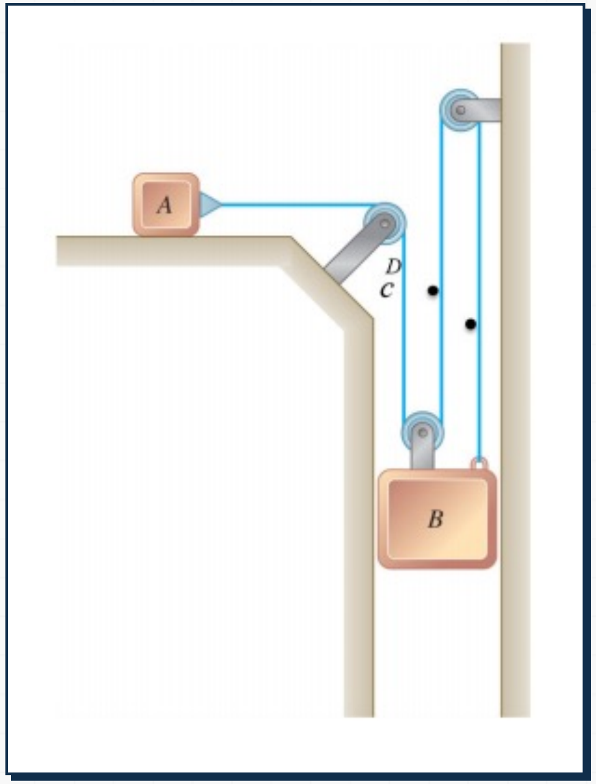
Reflect and Think

In this case, the relationship we needed was not between position coordinates, but between changes in position coordinates at two different times.

The key step is to clearly define your position vectors. This is a two degree-of-freedom system, because two coordinates are required to completely describe it.

Group Problem Solving

- Slider block A moves to the left with a constant velocity of $6m/s$.



- Determine
  - the velocity of block B.

GIST

Group Problem Solving

- **Slider block A moves to the left with a constant velocity of $6m/s$.**
  - Strategy
    - Sketch your system and choose coordinate system.
    - Write out constraint equation.
    - Differentiate the constraint equation to get velocity.

References

- Beer, Johnston, Cornwell, Self, Sanghi, “Vector Mechanics for Engineers: Dynamics” 12th Edition, Chapter 11